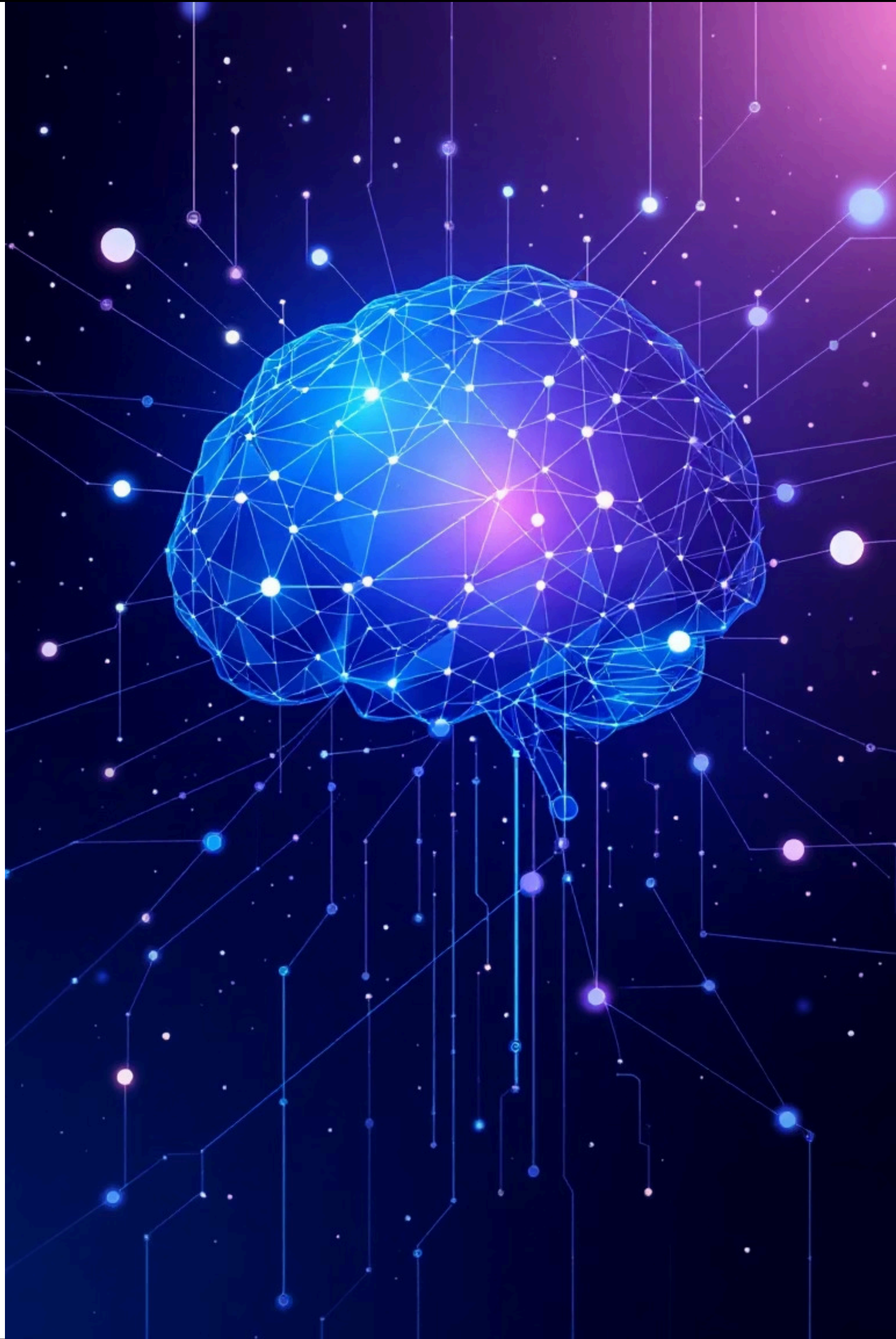
Module 03

Prompt Engineering Fundamentals

Designing Clear, Safe, and Reliable Prompts for GenAI Applications

GenAI, NLP & Prompt Engineering



MODULE 03

Prompt Engineering Fundamentals

Subject Focus

GenAI, Natural Language Processing, and Prompt Engineering

Module Goal

Designing Clear, Safe, and Reliable Prompts

Core Skill

Controlling AI Behavior Through Precise Instructions

This module builds on foundational concepts to teach you how to craft prompts that guide AI systems toward accurate, safe, and consistent outputs. You'll learn the essential structure of effective prompts and apply these principles to real-world tasks including summarization, data extraction, and content classification.

QUICK RECAP

Key Concepts from Module 02

01

Text to Numbers

AI systems must convert human language into numerical representations to process information

02

Tokenization

Text is split into discrete units called tokens, forming the basic building blocks of AI comprehension

03

Embeddings

Tokens are transformed into embeddings that capture semantic meaning in vector space

04

Similarity Measurement

Mathematical distances between embeddings enable AI to understand relationships and context

05

Controlled Responses

These foundational concepts enable precise control over AI behavior and output quality

Module Overview and Learning Goals

1

Quality Depends on Input

The quality and reliability of AI output is directly determined by the quality and precision of your input prompts

2

Prompts as Control Mechanisms

Effective prompts guide not just what the AI does, but how it responds, including format, tone, and scope

3

Design Principles

Learn the fundamental principles of clear, safe prompt design that minimize errors and ensure consistent results

4

Practical Applications

Apply prompt engineering techniques to common tasks: summarization, information extraction, and content classification

What Is a Prompt?

A prompt is fundamentally an instruction, not merely a question. It serves as the primary interface for controlling AI behavior, defining the task, establishing boundaries, and shaping the output format and tone.

Effective prompts tell the AI system **what to do**, **what information to use**, and **how to respond**. This precision is critical for reducing errors, preventing hallucinations, and ensuring reliable, repeatable results across different use cases.

- ## Task Definition

  Specifies exactly what action the AI should perform

- ## Scope Control

  Establishes boundaries and constraints for the response

- ## Format Specification

  Defines how the output should be structured and presented

- ## Tone Guidance

  Sets the voice and style for the AI's communication

Key Insight: Clear prompts reduce errors and hallucination by providing explicit guidance that eliminates ambiguity and guesswork.

Weak vs Strong Prompts

The difference between weak and strong prompts often determines whether an AI system produces useful results or generates confusion. Vague prompts lead to unpredictable outputs—responses that may be too long, irrelevant, or completely incorrect. Strong prompts specify clear rules, constraints, and expected formats.

Type	Weak Prompt Example	Strong Prompt Example
Summarization	"Tell me about climate change"	"Summarize the three main causes of climate change in 50 words or less, using bullet points"
Data Extraction	"Find information in this document"	"Extract all dates, monetary amounts, and company names from this contract. Return as JSON with keys: dates, amounts, companies"
Classification	"What kind of email is this?"	"Classify this email as one of: urgent, informational, spam, or action-required. Respond with only the category label"
Content Generation	"Write something about productivity"	"Write a 3-paragraph blog intro about time management for remote workers. Use a professional but friendly tone. Include one actionable tip in each paragraph"

Notice how strong prompts eliminate ambiguity by specifying output length, format structure, and exact requirements. This precision dramatically improves accuracy and creates consistency across multiple queries.

Why Prompt Engineering Is Needed

Repeatable Quality

Ensures consistent, high-quality output across different queries and use cases, making AI systems reliable for production environments

Reduces Hallucinations

Minimizes fabricated or incorrect answers by providing clear boundaries and explicit instructions about acceptable responses

Controls Length and Format

Specifies exact output requirements, preventing overly verbose responses or incorrect structural formatting

Enforces Safety Rules

Implements refusal protocols and safety guardrails that protect users from harmful, biased, or inappropriate content

Without proper prompt engineering, AI systems become unpredictable black boxes. Well-designed prompts transform AI into reliable, controlled tools that organizations can trust for critical business processes.

Core Structure of a High-Quality Prompt

Every effective prompt is built on three foundational layers that work together to guide AI behavior. Understanding and implementing these layers is essential for creating prompts that consistently deliver accurate, relevant, and appropriately formatted responses.

1

Layer 1: Instruction

Defines the specific task to perform with precision and clarity. Examples: "Summarize the following text," "Extract all product names," "Classify this document"

2

Layer 2: Context

Establishes rules, constraints, and boundaries. Includes information about data sources, assumptions to avoid, length limits, and quality requirements

3

Layer 3: Output Format

Specifies exactly how the answer should be structured and presented. Examples: bullet points, JSON schema, paragraph format, or specific field names

Critical Point: All three layers working together prevent confusion and ensure the AI understands not just *what* to do, but *how* and *within what constraints*.

Instruction, Context, and Output Layers

Let's examine each layer in detail to understand how they function individually and collectively to create powerful, reliable prompts.

1

Instruction Layer

Purpose: Defines the task with absolute precision

- Use action verbs: summarize, extract, classify, generate, analyze
- Be specific about scope: "Summarize the key findings" not "Tell me about this"
- State the desired outcome explicitly

Example: "Extract the customer name, order total, and delivery date from the email below"

2

Context Layer

Purpose: Limits assumptions and establishes boundaries

- Define what information to use and what to ignore
- Specify length constraints and quality thresholds
- Include domain-specific rules or terminology
- State what the AI should NOT do

Example: "Use only information from the document. Do not make assumptions about missing data. Limit response to 100 words"

3

Output Format Layer

Purpose: Controls structure and readability

- Specify exact format: bullets, JSON, table, paragraph
- Define field names for structured data
- Set requirements for citations or sources
- Establish tone and style guidelines

Example: "Return as JSON with keys: customer_name, order_total, delivery_date. Use ISO 8601 format for dates"

Zero-Shot vs Few-Shot Prompting

Zero-Shot Prompting

Provides instruction only, without examples. The AI must infer the desired behavior from the instruction alone.

When to use: Simple, well-defined tasks that don't require specific formatting or domain knowledge

```
Task: Classify this email as urgent or not urgent.

Email: "The server is down and customers can't access the system."

Classification:
```

Few-Shot Prompting

Provides instruction plus 2-5 examples that demonstrate the desired pattern, format, and quality level.

When to use: Complex tasks requiring specific formatting, nuanced understanding, or consistent style

```
Classify emails as urgent or not urgent.

Example 1:
Email: "Server down, customers affected"
Classification: urgent

Example 2:
Email: "Monthly newsletter attached"
Classification: not urgent

Email: "Payment system not processing"
Classification:
```

Guardrails, Refusals, and No-Hallucination Rules

Robust AI systems must know when to refuse requests and never invent information. Implementing clear safety rules protects users, maintains trust, and ensures institutional compliance with ethical standards.

Medical Advice

AI must refuse to diagnose conditions or prescribe treatments

"I cannot provide medical diagnosis. Please consult a healthcare professional"

Legal Guidance

Cannot offer specific legal advice or represent attorney services

"I cannot provide legal advice. Consult a licensed attorney for your situation"

Financial Planning

Must decline personalized investment recommendations or financial planning

"I cannot offer personalized financial advice. Speak with a certified financial advisor"

Missing Information

When data is unavailable, clearly state limitations instead of guessing

"The document does not contain delivery date information. I cannot provide this data"

No-Hallucination Principles

- Never invent facts, dates, or statistics not present in source material
- Clearly distinguish between provided information and general knowledge
- When uncertain, explicitly acknowledge uncertainty
- Provide confidence indicators when appropriate

Trust Building: Clear refusal messages improve user trust by demonstrating the system knows its limitations and prioritizes accuracy over completeness.

Structured Output and Prompt Iteration

Free-text responses are difficult to verify, parse, and integrate into automated workflows. Structured output formats like JSON enable reliable data extraction and systematic quality control.

Why Structured Output Matters

Machine Readability

Structured formats enable automated processing and integration with other systems

Validation

JSON schemas allow verification that all required fields are present and correctly formatted

Consistency

Same structure across all responses simplifies downstream processing and error handling

Testing

Structured output enables systematic quality testing and performance measurement

JSON Schema Example

```json
{
  "customer_name": "John Smith",
  "order_total": 149.99,
  "delivery_date": "2024-02-15",
  "items": [
    {
      "product": "Laptop",
      "quantity": 1,
      "price": 149.99
    }
  ],
  "status": "confirmed"
}
```

Ethical Prompting and In-Class Exercise

Ethical prompt design ensures AI systems respect user privacy, avoid perpetuating bias, and prevent misuse. These principles must be embedded at the prompt level, not just in model training.

No Harm

Refuse requests that could cause physical, psychological, or financial harm to individuals or groups

Bias Prevention

Avoid prompts that could produce discriminatory outputs based on protected characteristics

Data Privacy

Never request or expose personally identifiable information without explicit consent

No Plagiarism

Prevent direct copying of copyrighted material or academic dishonesty

No Impersonation

Refuse to impersonate real individuals or generate deceptive content

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In-Class Exercise (15 minutes)

Task: Write a complete prompt for summarizing customer feedback emails

Requirements:

- Include all three layers: instruction, context, and output format
- Add safety rules for handling sensitive customer data
- Specify structured JSON output with fields: sentiment, key_issues, priority_level
- Include a no-hallucination rule for missing information

Deliverable: One complete prompt ready for testing with sample data

Debrief, Homework, and Key Takeaways

Debrief Discussion

Why Clarity Matters

Discuss how prompt clarity directly impacts output quality, reliability, and user trust in production systems

Strong Prompt Example

Share one example from your exercise that demonstrates effective use of instruction, context, and output layers

Unsafe Prompt Analysis

Identify one unsafe prompt scenario and explain why it fails to meet ethical or safety standards

Homework Assignment

1. **Rewrite Weak Prompts**
   Transform 3 vague prompts into strong prompts with clear instruction, context, and format layers
2. **Add Safety Rules**
   Include refusal protocols and no-hallucination rules for each rewritten prompt
3. **JSON Output**
   Design a structured JSON schema for at least one data extraction task

Key Takeaways

- Prompts are instructions that control AI task, scope, format, and tone
- Strong prompts have three layers: instruction, context, and output format
- Few-shot prompting improves accuracy through examples
- Safety guardrails and refusal rules protect users and maintain trust
- Structured output enables validation, automation, and quality control
- Prompt iteration is essential for handling edge cases and improving reliability

Thank You

Questions?

NEXT MODULE

MODULE 04 – Retrieval Basics: Simple Search Using NumPy & Cosine Similarity